DIFFICULT INTERVIEW QUESTIONS

Q : If there are un-equal channels in EDT, how to calculate compression ratio? Is there any limitations of having unequal channels in EDT?

Ans :- no. of internal chains = 15
no. of i/p channel = 2
no. of o/p channel = 1

compression ratio = 15/[(2+1)/2]
= 15/1.5
≃ 10x

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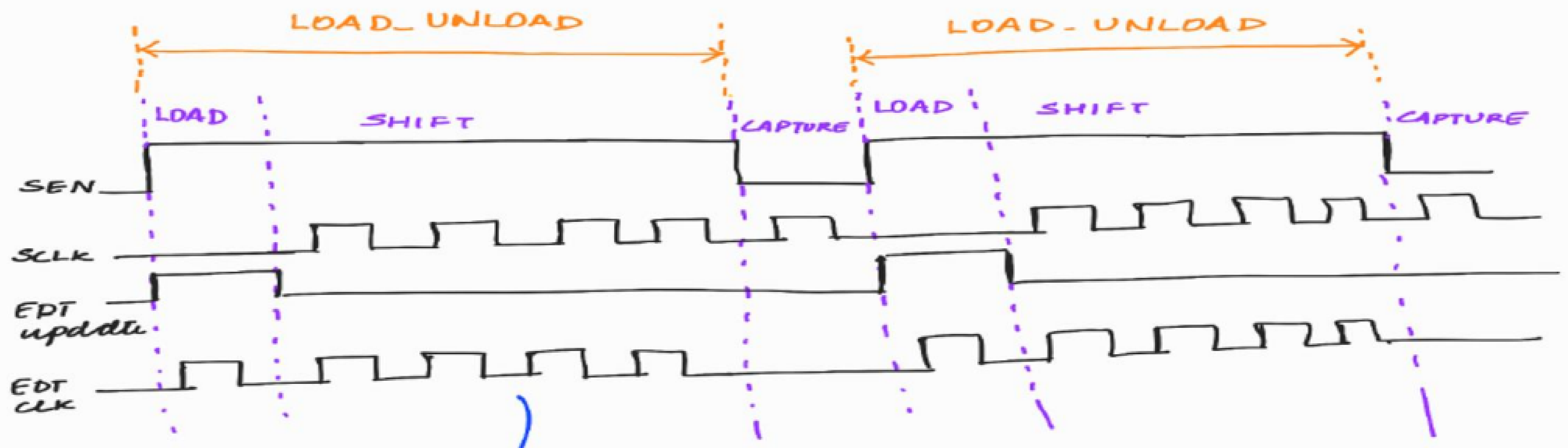
limitations of having unequal channels:
in uncompressed mode (bypass mode), it will utilize only min of 2.

Eg :- i/p channels 3
o/p channels 2
Bypass = min (3, 2) = 2

Eg:- i/p channels 1
o/p channels 2
Bypass = min (1,2) = 1

Q2.



no. of shift cycles per pattern } = no. of ffs in the scan chain + masking bits + no. of initialization cycles

↳ ceil (decomp size / no. of channels)

Q(a) Why Edt.clk is not pulsed during capture?

Ans:- Capture phase is to capture the response of the combo logic. In EDT, there is no significance of this phase because EDT is used to decompress and shift appropriate pattern into the scan chain and to compress the values shifted out by the scan chains.

So edt.clk is not pulsed during cap.

Q(b) Why SCLK (Scan clk) is not pulsed during load phase?

Ans :- Note :- For understanding, refer to illus (2) and (3) waveform

Consider the second shift phase.
During the second shift phase, the captured value of P1 will be shifted out and simultaneously P2 will be shifted in.

Suppose SCLK (slow freq clk that goes to SFFs) is pulsed during load phase, then the cap value of SFFs will start to get shifted out in the load phase itself.

But cap value of SFFs should start to get shifted out in the shift phase.

So SCLK is not pulsed during load phase.